QUALITY ANALYSIS OF BASMATI RICE USING IMAGE PROCESSING TECHNIQUES

A PROJECT REPORT SUBMITTED BY

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(S / 18 / 338)

For the partial fulfillment for the requirement for the course of

CS 314 Image Processing Practical

To the **Department of Statistics and Computer Science** of the

FACULTY OF SCIENCE
UNIVERSITY OF PERADENIYA
SRI LANKA
2022

DECLARATION

I do hereby declare that the work reported in this project report was exclusively carried out by me under the supervision of Prof. Amalka Pinidiyaarachchi. It describes the results of my own independent work except where due reference has been made in the text. No part of this project report has been submitted earlier or concurrently for the same or any other degree.

Date: ……. ……

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ABSTRACT

Rice is the staple food for 80% of the population in the world. In Sri Lanka, nearly half of the total daily calorie intake of an average person is obtained through rice. Among different rice types Basmati rice which is also termed the king of rice is very popular not only because of its taste, grain length and aroma but also when compared to regular plain grain rice, Basmati exhibits rich fiber content. When the crop turns to a golden color harvesting of basmati begins. Then the harvested crop is milled. After milling, rice is sent to the market to get distributed among consumers. The impurity content is an important index to distinguish the rice quality, which directly impacts its market competitiveness. The presence of impurities like stones, weed seeds, chaff, and damaged broken seeds leads to the poor quality of rice. Based on the rice grains (full, half, broken) basmati rice is graded in the market as the grain size is an important factor when considering the quality of basmati rice. Rice grading is a process of sorting rice and assigning it into grades based on its properties. At present the traditional rice impurity manual inspection is very time-consuming and inefficient. So the main objective of this project is to analyze the quality of basmati rice and the grading of rice based on features such as the major axis, minor axis, aspect ratio and the chalky percentage of the rice kernel using image processing techniques.

ACKNOWLEDGEMENTS

I would like to extend my sincere gratitude to the following without whose support and guidance throughout, this project would not have been possible. First, I would like to thank Dr. Amalka Pinidiyaarachchi for providing theoretical and practical knowledge in the field of digital image processing. I would like to extend my sincere gratitude to Mr. Rochana Obadage and Mr.Kansaji Kotuwage for providing the necessary practical knowledge and guidance throughout the project. I would like to extend my sincere gratitude to Dr. Sachith Abeysundara, head of the department of Statistics and Computer Science for providing the facilities to make this project successful. Finally, I would like to express my gratitude to my family and colleagues for their kind cooperation and encouragement which helped me in the completion of this project.

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LIST OF ABBREVIATIONS

OpenCV – Open Source Computer Vision Library

Numpy – Numerical Python

IRRI – International Rice Research Institute

CHAPTER 01

1.1. INTRODUCTION

Rice is the staple food for 80% of the population in Asian countries. In Sri Lanka the main calorie intake of people is rice. Among all the other varieties of rice is Basmati rice, renowned for its unique fragrance, elongated grains, and rich fiber content. The journey of Basmati rice form the paddy fields to dinner tables involves a long process. It involves the milling process which is removing the husk and bran layers to obtain the edible grain. This process significantly affects the appearance, and texture characteristics. When milled rice comes to the market, commercially it is graded on many factors such as length of the grain, moisture content, availability of broken rice and availability of impurities, the amount of chalky present in the seed. Rice quality depends on both physical and chemical characteristics. Grain size and shape, chalkiness, whiteness, milling degree, bulk density and moisture content are some physical characteristics while amylose content, gelatinization temperature and gel consistency are chemical characteristics of rice. The quality of the grains is referred to the every aspect of the profit of supply and marketing. So the milling process has a high impact on the quality of the rice. Traditionally the milling process is done manually which is very inefficient and time consuming. It consumes lot of man power, time and money.

1.2. PROBLEM STATEMENT

The quality of basmati rice grain is an important factor when grading the rice commercially. But traditionally this grading is done manually, which cost more man power time and cost which is also inefficient. In traditional method people use “Vernier caliper” to measure the length of basmati rice and other equipment like grain shape tester, micrometer to measure length and breadth of single grain at a time. The outcome of this analysis is also relative, time consuming, having variable results and costly. And also the demand for rice is increasing tremendously hence producing and sorting of rice becoming more faster than usual requirements. So it requires high degree of accuracy to satisfy customers need and to overcome limitations of manual inspection.

1.3. OBJECTIVE

The main objective of this study is to design and develop an image-processing system for analyzing the quality of basmati rice grains. This mini project aims to use image processing techniques to analyze basmati rice grains and classify the grains according to the length of rice grains, aspect ratio of rice grain and to measure the chalky percentage in rice grains.

1.4. APPROACH

For this project, I have used image processing techniques through the OpenCV library and Python programming language.

CHAPTER 02

RELATED WORK

The agricultural industry is oldest and most widespread industry in the world. Rice is grown in many countries, as it is the most important food crop of the emerging world. In Sri Lanka still the quality inspection is mainly done in manual process.

People use manual inspection method to define the quality of rice. The traditional methods used for grain size and shape measurements are dial micrometer, Vernier caliper, grain shape tester which measures a single grain at a time.

Numerous attempts and a lot of research has been done on quality analysis of rice. These researches are mainly focusing the grain length and the grain breadth.

They have suggested the way of grading based on these physical qualities. However in these research and other attempts they have not focused on the factor of chalky in the rice grain, br.

Property of chalky is also a main factor that affect the quality of rice. Chalky grain is grain at least half of which is milky white in color and brittle in nature. And also the parameter of number of broken seeds is also an essential factor in rice quality analysis which is not very well emphasized in the research works.

However, some approaches have employed ensemble machine learning and use of neural network pattern recognition system for grading rice granules which is also based on the length of rice grains.

CHAPTER 03

METHODOLOGY

3.1. IMAGE PROCESSING PIPELINE

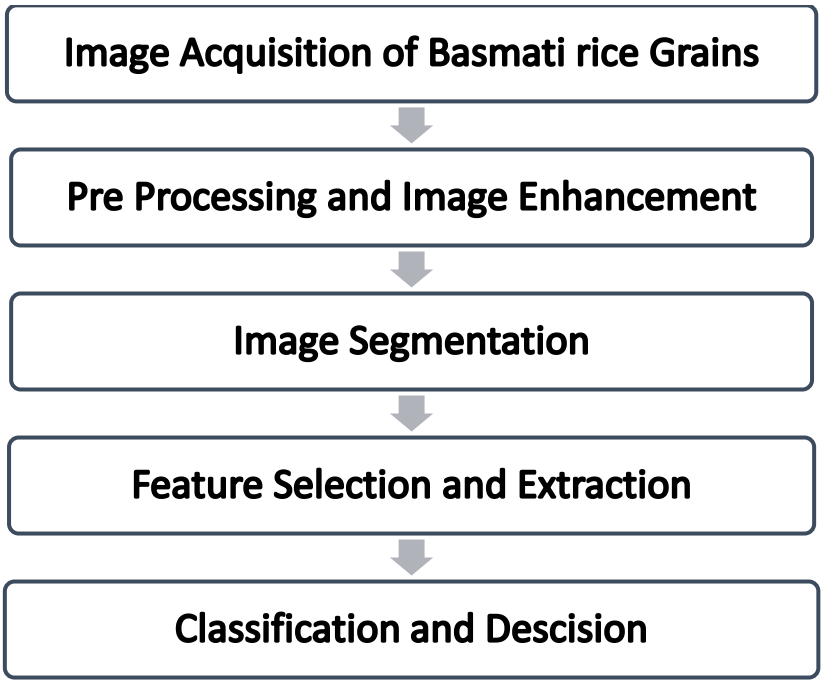


Figure 1: Flow Chart of the methodology

3.2. SETTING UP THE OPEN CV ENVIRONMENT

This project was developed using Python 3.11.2

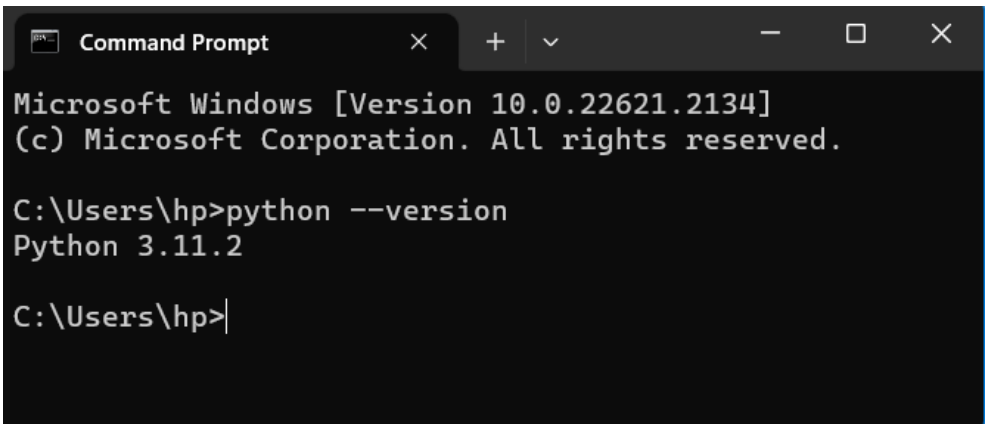


Figure 2: Python Environment

First, the required libraries were imported. The imported libraries are;

- OpenCV
- Numpy
- Matplotlib

```
quality_analysis_of_basmati.py > classification_ratio
1    import cv2
2    import numpy as np
3    import matplotlib.pyplot as plt
4
```

Figure 3: List of libraries used

3.3. IMAGE ACQUISITION

For this project I have created my own dataset of Basmati rice. Three samples of Basmati rice were collected. First sample taken from the store, second and third sample was collected from a rice mill where second sample obtained from a traditional milling machine and the third sample was obtained from a advanced rice milling machine. The images are captured by a smart phone. To collect the images, the smartphone was placed in a location parallel to the plane. A black background is used.



Figure 4 : Image Acquisition 01



Figure 4.1 : Image Acquisition 02

Captured Images saved in the format of .jpeg , .jpg, .png



Figure 4.2: Sample Image 1



Figure 4.3: Sample Image 2



Figure 4.4: Sample Image 3



Figure 4.5 : Sample Image 4

3.4. PROCEDURE

- First the images were read using imread() function that is available in OpenCV library, and then images were converted into grayscale and rgb scale using cvtColor() function in OpenCV library.

```
#loading the image
img_bgr = cv2.imread("Images/img_01_pure.jpg", 1)
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
img = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)
```

3.4.1. IMAGE PROCESSING

- To reduce the impact of noise in the images noise reduction techniques were applied.
- Then thresholding was applied to segment the rice grains from the black background. Here I have used the threshold value of 160.

```
median_filtered = cv2.medianBlur(img, 5)
ret, thresh = cv2.threshold(median_filtered, 160, 255,
cv2.THRESH_BINARY)
```

3.4.2. APPLYING MORPHOLOGICAL OPERATIONS

- A kernel was defined of size (3,3).
- Erosion was applied to the images to separate the touching features of rice grains without losing the integrity of a single feature.
- Dilation process follows erosion process. The objective of applying dilation is to grow the eroded features to their original shape without rejoining the seperated features.

```
kernel = np.ones((3,3), np.float32)/9
kernel2 = cv2.getStructuringElement(cv2.MORPH_ELLIPSE,(3,3))
erosion = cv2.erode(dst, kernel2, iterations=3)
dilation = cv2.dilate(erosion, kernel2, iterations=1)
```

3.4.3 EDGE DETECTION

- Edge detection is applied so that it helps to find out the region of boundaries of rice grains. Here I have used the Canny edge detection algorithm to detect edges.

```
edges = cv2.Canny(dilation, 100, 200)
```

3.4.4. OBJECT MEASUREMENT

- The main measurements that were obtained were the length of rice grain and the breadth of the rice grain.
- The number of rice grains in each sample was also calculated.
- Area of each rice grains also calculated.
- The contours are defined using findContours() function and the contours were drawn using drawContours() function.

```
contours, hierarchy = cv2.findContours(erosion, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)
num_seeds = len(contours)
print("No of rice grains = ", num_seeds)
contour_draw = cv2.drawContours(dilation_color, contours, -1, (0,255,0), 2)

major_axis_length = []
minor_axis_length = []
Aspect_ratio_of_seeds = []

for idx, contour in enumerate(contours):
        seed_number = num_seeds - idx
       x, y, w, h = cv2.boundingRect(contour)
       aspect_ratio = float(w)/h
       if(aspect_ratio < 1):
               aspect_ratio = 1/ aspect_ratio
      major_axis = w
      minor_axis = h

        print("Seed Number: ", seed_number, " | ",
              "Major axis length: ", major_axis, "|",
              "Minor axis length: ", minor_axis, "|",
              "Aspect Ratio: ",aspect_ratio, "|")

        major_axis_length.append(major_axis)
        minor_axis_length.append(minor_axis)

        cv2.putText(contour_draw, str(seed_number), (int(center[0]),
int(center[1])),cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 0, 0), 2)
```

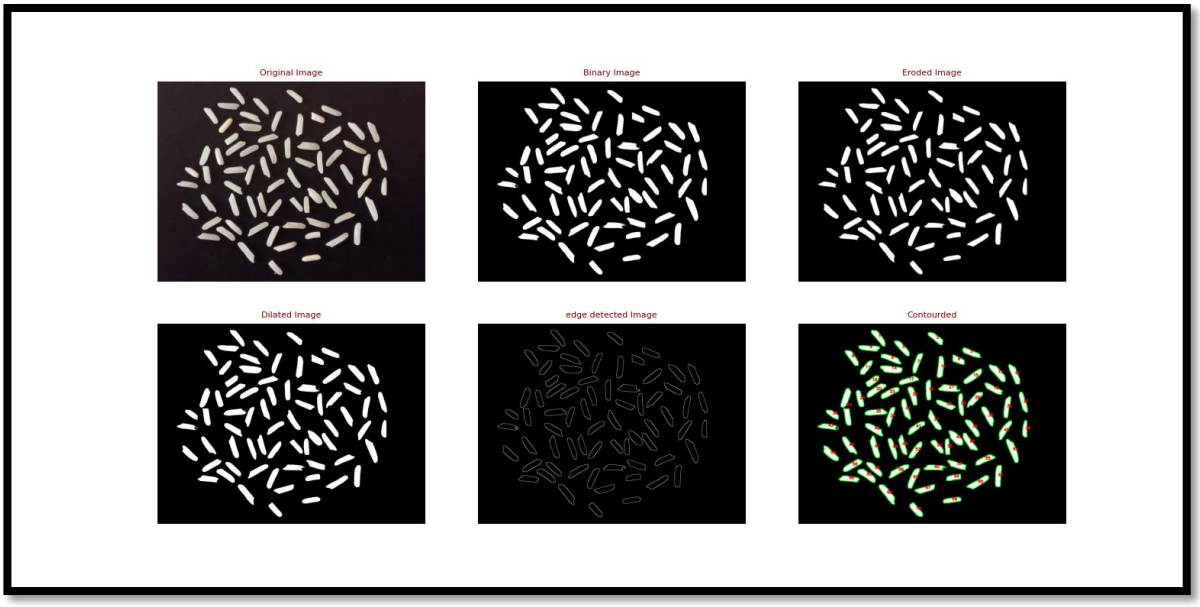


Figure 5: Output images of the steps to detect and count rice grains

```
PS F:\Third Year\CS 314 PROJECT\CODE BANK\Code CS 314> & "C:/Program Files/Python311/python.exe" "f:/Third Year/CS 314 PROJECT/CODE BANK/Code CS 314/quality_analysis_of_ba
smati.py"
No of rice grains =  70
Seed Number:  70  |  Major axis length:  67.10263061523438 | Minor axis length:  22.139318466186523 | Aspect Ratio:  3.030925758519645 |
Seed Number:  69  |  Major axis length:  75.09298706054688 | Minor axis length:  16.522178649902344 | Aspect Ratio:  4.5449809405729145 |
Seed Number:  68  |  Major axis length:  85.98969268798828 | Minor axis length:  18.654693603515625 | Aspect Ratio:  4.609547308339755 |
Seed Number:  67  |  Major axis length:  105.47045135498047 | Minor axis length:  22.993438720703125 | Aspect Ratio:  4.586980339744296 |
Seed Number:  66  |  Major axis length:  104.10112762451172 | Minor axis length:  19.942068099975586 | Aspect Ratio:  5.220177120177379 |
Seed Number:  65  |  Major axis length:  79.74593353271484 | Minor axis length:  19.325462341308594 | Aspect Ratio:  4.1264696349466465 |
Seed Number:  64  |  Major axis length:  65.9420166015625 | Minor axis length:  18.1729679107666 | Aspect Ratio:  3.628577177121138 |
Seed Number:  63  |  Major axis length:  92.07289123535156 | Minor axis length:  21.56730842590332 | Aspect Ratio:  4.26909512383881 |
Seed Number:  62  |  Major axis length:  99.94580841064453 | Minor axis length:  19.637752532958984 | Aspect Ratio:  5.089472853011091 |
Seed Number:  61  |  Major axis length:  94.45138549804688 | Minor axis length:  21.191431045532227 | Aspect Ratio:  4.457055556800634 |
Seed Number:  60  |  Major axis length:  75.51390838623047 | Minor axis length:  21.575658798217773 | Aspect Ratio:  3.4999584064829663 |
Seed Number:  59  |  Major axis length:  97.468994140625 | Minor axis length:  19.318952560424805 | Aspect Ratio:  5.045252522659632 |
Seed Number:  58  |  Major axis length:  76.59907531738281 | Minor axis length:  19.34750747680664 | Aspect Ratio:  3.959118527759096 |
Seed Number:  57  |  Major axis length:  84.2599105834961 | Minor axis length:  19.830625534057617 | Aspect Ratio:  4.248978956250573 |
Seed Number:  56  |  Major axis length:  88.3000259399414 | Minor axis length:  19.925600051879883 | Aspect Ratio:  4.431486414965492 |
Seed Number:  55  |  Major axis length:  85.71647644042969 | Minor axis length:  18.86551284790039 | Aspect Ratio:  4.543554004150455 |
Seed Number:  54  |  Major axis length:  75.98128509521484 | Minor axis length:  24.333112716674805 | Aspect Ratio:  3.122546875934495 |
Seed Number:  53  |  Major axis length:  85.31256103515625 | Minor axis length:  18.615537643432617 | Aspect Ratio:  4.582868497770931 |
Seed Number:  52  |  Major axis length:  65.42216491699219 | Minor axis length:  18.99732208251953 | Aspect Ratio:  3.443757211296148 |
Seed Number:  51  |  Major axis length:  104.14959716796875 | Minor axis length:  22.761186599731445 | Aspect Ratio:  4.575754287309326 |
Seed Number:  50  |  Major axis length:  68.2221450805664 | Minor axis length:  16.309720993041992 | Aspect Ratio:  4.182913068204609 |
Seed Number:  49  |  Major axis length:  95.64606475830078 | Minor axis length:  16.08213233947754 | Aspect Ratio:  5.947349688418746 |
Seed Number:  48  |  Major axis length:  81.81828308105469 | Minor axis length:  20.896865844726562 | Aspect Ratio:  3.9153375290343826 |
Seed Number:  47  |  Major axis length:  91.8093490600586 | Minor axis length:  20.49503517150879 | Aspect Ratio:  4.479589729501296 |
Seed Number:  46  |  Major axis length:  80.6683578491211 | Minor axis length:  18.870637893676758 | Aspect Ratio:  4.274808212824206 |
Seed Number:  45  |  Major axis length:  92.579833984375 | Minor axis length:  20.50023078918457 | Aspect Ratio:  4.516038621048984 |
Seed Number:  44  |  Major axis length:  68.52198791503906 | Minor axis length:  27.516725540161133 | Aspect Ratio:  2.4901941117605015 |
Seed Number:  43  |  Major axis length:  69.4209213256836 | Minor axis length:  17.59610366821289 | Aspect Ratio:  3.9452439377867212 |
```

Figure 5.1: Output Results of measurements

3.4.5. OBJECT CLASSIFICATION

Classification and grading of rice grains were done in this step. The classification is done based on the length of rice grains and the aspect ratio of the rice grain, based on the chalky content, based on broken seed content and based on impurity content in a given rice sample. Here for all the classifications I have used the standards provided by International Rice Research Institute.

3.4.5. 1.Based on Grain length (Major Axis Length)

The length and width of a rice grain are important attributes that determine the class of the rice. There are four main classes of basmati rice, based on grain length. They are extra-long, long, medium and short. The length and width is used internationally to describe the shape and class of variety. The grain length gives the uniformity to the sample.

Class	**Major Axis length (pixels)**
Extra-long grain rice	Above 42
Long grain rice	42 - 44
Medium grain rice	30 - 42
Short grain rice	26 - 30

Table 01: Standards for major axis length

- Here I first defined rice class varieties as Extra-long grain, long grain, medium grain, short grain, and broken gain. And then defined count variables to count the number of rice grains fall under each section which satisfies the conditions

```
count_extra_long = 0
count_long = 0
count_medium = 0
count_short = 0
count_broken = 0

rice_varieties = {
    "Extra Long Grain": count_extra_long,
    "Long Grain": count_long,
    "Medium Grain": count_medium,
    "Short Grain": count_short,
    "Broken Grain": count_broken
}
```

- Then created a method by passing the major axis length as the parameter to get the count of each rice grain falls in to each respective class.

```
def classification_length(axis_length):
    global count_extra_long, count_long, count_medium, count_short,
count_broken
    length_class = ""
    if axis_length >= 85:
        length_class = "Extra Long Grain"
        count_extra_long += 1
        print("count_extra_long:", count_extra_long)
    elif(axis_length < 85 and axis_length >= 60):
        length_class = "Long Grain"
        count_long += 1
        print("count_long:", count_long)
    elif(axis_length < 60 and axis_length >= 50):
        length_class = "Medium  Grain"
        count_medium += 1
        print("count_medium:", count_medium)
    elif (axis_length < 50 and axis_length >= 30):
        length_class = "Short Grain"
        count_short += 1
        print("count_short:", count_short)
    else:
        length_class = "Broken Grain"
        count_broken += 1
        print("count_broken:", count_broken)

    cv2.putText(dilation_color, "Extra Long: " + str(count_extra_long),
(10, 20), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)
    cv2.putText(dilation_color, "Long: " + str(count_long), (10, 40),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)
    cv2.putText(dilation_color, "Medium: " + str(count_medium), (10,
60), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)
    cv2.putText(dilation_color, "Short: " + str(count_short), (10, 80),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)
    cv2.putText(dilation_color, "Broken: " + str(count_broken), (10,
100), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)
    return length_class
```

- Then I have got the sum of the counts of rice grains that falls on the given classes separately and calculated the percentage content of each class.

```
 Print the counts of each rice variety
print("No of rice grains = ", len(contours))
for variety, count in rice_varieties.items():
    print(f"Count of {variety}: {count}")
```

```
total_seeds = len(contours)
percentage_extra_long_grain = (count_extra_long / total_seeds) * 100
percentage_long_grain = (count_long / total_seeds) * 100
percentage_medium_grain = (count_medium / total_seeds) * 100
percentage_short_grain = (count_short / total_seeds) * 100
percentage_broken_grain = (count_broken / total_seeds) * 100
# Print  the analysis
print("Analysis based on the major axis length of the rice granules:")
# Display the analysis based on major axis length
print("{:.2f}% of Very High Quality Rice".format
(percentage_extra_long_grain))
print("{:.2f}% of High Quality Rice".format(percentage_long_grain))
print("{:.2f}% of  Average Quality Rice".format(percentage_medium_grain))
print("{:.2f}% of Low Quality Rice".format(percentage_short_grain))
print("{:.2f}% of Very Low Quality Rice".format(percentage_broken_grain)
```

- **Graphical Representations**

Histograms were used to show the distribution of major axis length and minor axis length

```
#Major axis distribution
plt.hist(major_axis_length, bins=100, color='green', alpha =0.7, edgecolor =
'black')
plt.xlabel("Major Axis Length")
plt.ylabel('frequency')
plt.title("Distribution of major axis lengths", color = "maroon",fontsize =
8)
plt.grid(True)

#Minor axis distribution
plt.hist(minor_axis_length, bins=100, color='blue', alpha =0.7,edgecolor =
'black')
plt.xlabel("Minor Axis Length")
plt.ylabel('frequency')
plt.title("Distribution of minor axis lengths", color = "maroon", fontsize =
8)
plt.grid(True)
```

Pie chart was used to show the quality analysis of Basmati rice based on major axis length

```
quality_categories = ["Very High Quality", "High Quality", "Average
Quality", "Low Quality", "Very Low Quality"]
quality_percentages = [percentage_extra_long_grain, percentage_long_grain,
percentage_medium_grain, percentage_short_grain,percentage_broken_grain]
plt.pie(quality_percentages, labels=quality_categories, autopct='%.2f%%',
colors=['green', 'blue', 'purple','yellow','red'])
plt.title("Quality Percentage of Rice Sample", color = "maroon")
```

3.4.5.2.Based Grain Shape

-

Scale	Shape	Length/ width ratio
1	Slender	Over 3.0
3	Medium	2.1 - 3.0
5	Bold	1.1 – 2.0
9	Round	1.0 or less

Table 02: Standards for Grain shape/ Aspect ratio

$$Aspect\ ratio = \frac{length\ of\ rice\ * 100}{Width\ of\ rice}$$

  - Next by considering the shape of the rice grains such as Slender, Medium, Bold and Round analysis of rice quality is done.
  - For that First defined shape varieties and then created count variables to store the count of rice grains falls into each class.

```
shape_varieties = {
    "Extra Long Grain": count_slender,
    "Medium Grain": count_medium,
    "Short Grain": count_bold,
    "Broken Grain": count_round
}
count_slender = 0
count_medium = 0
count_bold = 0
count_round = 0
```

- Based on the equation stated above the aspect ratios were calculated
- Then a function was created to group the rice grains based on the conditions stated above in the table to classify the rice grains in the image to the respective classes and the count of each class was obtained.

```
def classification_ratio(ratio):
    global count_slender, count_medium, count_bold, count_round
    ratio_class = ""
    if ratio >= 3:
        ratio_class = "Slender Grain Shape"
        count_slender += 1
        print("Slender Shape count: ", count_slender)
    elif(ratio < 3 and ratio > 2.1):
        ratio_class = "Medium Grain Shape"
        count_medium += 1
        print("Medium Shape count: ", count_medium)
    elif(ratio <= 2.1 and ratio >= 1.1):
        ratio_class = "Bold Size Grain"
        count_bold += 1
        print("Bold Shape count: ", count_bold)
    else:
        ratio_class = "Round Grain"
        count_round += 1
        print("Slender Shape count: ", count_round)
    return ratio_class
```

- Then the percentage of amount of rice grains in each classes were calculated. And then based on the percentages analysis of rice kernels were done.

```
print("No of rice grains = ", len(contours))
for variety, count in shape_varieties.items():
    print(f"Count of {variety}: {count}")

total_seeds = len(contours)
percentage_slender_grain = (count_slender / total_seeds) * 100
percentage_medium_grain = (count_medium / total_seeds) * 100
percentage_bold_size_grain = (count_bold / total_seeds) * 100
percentage_round_grain = (count_round / total_seeds) * 100

# Print the calculated percentages
print("Percentage of Slender Grain:", percentage_slender_grain)
print("Percentage of Medium Grain:", percentage_medium_grain)
print("Percentage of Bold Size Grain:", percentage_bold_size_grain)
print("Percentage of Round Grain:", percentage_round_grain)

# Print  the analysis
print("Analysis based on the aspect ratio of the rice granules:")
```

```
# Display the analysis based on major axis length
 print("{:.2f}% of Very High Quality Rice".format(percentage_slender_grain))
print("{:.2f}% of High Quality Rice".format(percentage_medium_grain))
print("{:.2f}% of Average Quality Rice".format(percentage_bold_size_grain))
print("{:.2f}% of Low Quality Rice".format(percentage_round_grain))
```

Pie Chart was used to show the analysis based on the aspect ratio of a given rice sample.

```
counts = [percentage_slender_grain, percentage_medium_grain,
percentage_bold_size_grain, percentage_round_grain]
counts = [count_slender, count_medium, count_bold, count_round]
# Classification ratio labels
labels = ["Slender Grain", "Medium Grain", "Bold Size Grain", "Round Grain"]
# Colors for each category
colors = ['skyblue', 'lightgreen', 'lightcoral', 'lightsalmon']
explode = (0.1, 0, 0, 0)
plt.pie(counts, labels=labels, colors=colors, explode=explode,
autopct='%1.1f%%', shadow=True, startangle=140)
plt.title("Distribution of Classification Ratios", color = "maroon",
fontsize = 8)
```

3.4.5. 3.Based on Chalkiness



Figure 06: Chalky Rice Grain

If part of the milled rice kernel is opaque rather than translucent, it is often characterized as “Chalky”. Chalkiness downgrades milled rice. Excessive chalkiness is caused by interruption during the final stage of grain filling. Even though chalkiness disappears upon cooking and has no direct effect on cooking, excessive chalkiness downgrades the quality and reduces milling recovery.

For detecting chalky in the rice grain extended maxima operator was used. By using this operator we can close all minimum values in the image and highlight the high values. By that we can separate the chalky portion in images.

% of chalky degree (chalky area/ area)	**Quality**
>30%	Bad
<30%	Good

Table 03: Standards for chalky content

$$\%\ Chalky\ grain = \frac{Total\ chalky\ area\ of\ rice\ * 100}{Total\ area\ of\ rice}$$

- In chalky detection part several steps were followed to detect the chalky area of each rice kernel.
    - Step 01: Apply Morphology Opening: This is the dilation of erosion
    - Step 02: Apply Morphology Erosion and Reconstruction: To remove smallest particles in the image
    - Step 03: Apply Morphological Closing operation: Opening removes small objects while closing removes small holes.
    - Step 04: Apply extended maxima transform
    - Step 05: Superimposed regional maxima on the original image
    - Step 06: Compute the area of chalky

```
#Apply canny edge detector
edges = cv2.Canny(image, 100, 200)
# Create a structuring element for the extended maxima transform
kernel = np.ones((3, 3), np.uint8)
# Apply morphological reconstruction
reconstruction = cv2.dilate(edges, kernel)
# Apply extended maxima transform
extended_maxima = reconstruction - edges
# Calculate the total area of chalky regions
chalky_area = np.sum(extended_maxima == 255)
print("Chalky Area:", chalky_area)
```

- Next total area of the rice kernels were obtained

```
contours, _ = cv2.findContours(extended_maxima, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

# Calculate the total area of rice seeds
total_rice_area = 0
for idx, contour in enumerate(contours):
    contour_area = cv2.contourArea(contour)
    print(f"Area of Rice Seed {idx + 1}: {contour_area}")
    total_rice_area += contour_area

print("Total area of Rice Seeds:", total_rice_area)
```

- Then chalky percentage was calculated based on the above stated equation

```
chalky_percentage = (chalky_area / total_rice_area) *100
print("Chalky percentage: ", chalky_percentage)
```

- Finally based on the chalky percentage rice grains were graded as good or bad.

```
if(chalky_percentage > 30):
    quality = "Bad"
else:
    quality  = "Good"

print(f"Percentage of chalky area: {chalky_percentage:.2f}%")
print(f"Quality: {quality}")
```

3.4.5.4. Based on the amount of broken rice grains in a sample

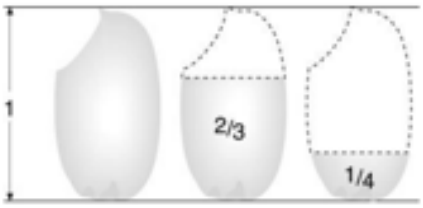


Figure 07: Broken Rice Grain

- Broken denoted pieces of kernels that are less than three fourth of a whole kernel. Pieces smaller than one fourth of the whole kernel are to be treated as fragments.
- The amount of broken rice grains in a rice sample also affects for the quality of basmati rice.
- To find the quality based on the broken rice seed content the following measurements given by International Rice Research Institute were used.

Quality Measurement	Broken Percentage (%)
Very Good Quality	<= 4.90
Good Quality	<= 19.75
Low Quality	<= 34.50
Very Bad Quality	<= 40.00

Table 04: Standards for broken seed content

- So to do that first area of each rice grain in the image was obtained.

```
for idx, contour in enumerate(contours):
    contour_area = cv2.contourArea(contour)
    print(f"Area of Rice Seed {idx + 1}: {contour_area}")
```

```
PS F:\Third Year\CS 314 PROJECT\CODE BANK\Code CS 314> & "C:/Program Files/Python311/python.exe" "f:/Third Year/CS 314 PROJECT/CODE BANK/Code CS 314/broken.py"
No of rice grains =  70
Area of Rice Seed 1: 1301.5
Area of Rice Seed 2: 1117.0
Area of Rice Seed 3: 1924.0
Area of Rice Seed 4: 1370.5
Area of Rice Seed 5: 1316.0
Area of Rice Seed 6: 1709.0
Area of Rice Seed 7: 1001.5
Area of Rice Seed 8: 1598.0
Area of Rice Seed 9: 1606.0
Area of Rice Seed 10: 1636.0
Area of Rice Seed 11: 1256.0
Area of Rice Seed 12: 1568.5
Area of Rice Seed 13: 1357.0
Area of Rice Seed 14: 1368.5
Area of Rice Seed 15: 1542.5
Area of Rice Seed 16: 1386.0
Area of Rice Seed 17: 1569.5
Area of Rice Seed 18: 1365.0
Area of Rice Seed 19: 1102.0
Area of Rice Seed 20: 1990.5
Area of Rice Seed 21: 969.5
Area of Rice Seed 22: 1326.5
Area of Rice Seed 23: 1503.0
Area of Rice Seed 24: 1644.0
Area of Rice Seed 25: 1324.0
Area of Rice Seed 26: 1517.0
Area of Rice Seed 27: 1676.5
Area of Rice Seed 28: 1156.0
Area of Rice Seed 29: 1477.5
Area of Rice Seed 30: 1335.0
Area of Rice Seed 31: 1477.5
Area of Rice Seed 32: 942.0
Area of Rice Seed 33: 1434.5
Area of Rice Seed 34: 1533.5
Area of Rice Seed 35: 927.5
Area of Rice Seed 36: 1507.0
Area of Rice Seed 37: 1249.5
Area of Rice Seed 38: 1578.5
```

Figure 8: Area of each rice grain in a sample

- And then mean weight of the sample was calculated.

```
mean_area = np.mean(areas)
```

- Then threshold area for broken rice was calculated using the fact that a kernel with area that are less than three fourth of whole grain.

```
three_fourths_mean_area = 0.75 * mean_area
```

- Two count variables were initialized and then finally count of broken seeds and healthy seeds were obtained.

```
# Define classification thresholds
broken_threshold = 0.75 * mean_area
print(broken_threshold)
fragment_threshold = 0.25 * mean_area
broken_count = 0
healthy_count = 0

for contour, area in zip(contours, areas):
    if area <= broken_threshold:
        cv2.drawContours(dilation_color, [contour], -1, (255, 0, 0), 2)  # Blue contour for fragment
        broken_count+=1
    else:
        cv2.drawContours(dilation_color, [contour], -1, (0, 255, 0), 2)  # Green contour for quality
        healthy_count+=1

print("Broken count: ", broken_count)
print("Healthy count: ", healthy_count)
```

```
def broken_quality(broken_percentage):
    rice_quality = ""
    if (broken_percentage <= 4.90):
        rice_quality =  "Very Good Quality"
    elif(4.90 <broken_percentage <= 19.75 ):
        rice_quality =  "Good Quality"
    elif(19.75 <broken_percentage <= 34.50):
        rice_quality =  "Low Quality"
    else:
        rice_quality =  "Very Bad Quality"
    return rice_quality

print("Quality of Rice : ", broken_quality(broken_percentage))
```

Pie chart was used to display the analysis based on broken seed content of a given rice sample.

```
labels = ['Broken', 'Healthy']
sizes = [broken_count, healthy_count]
colors = ['red', 'green']
explode = (0.1, 0)
plt.pie(sizes, explode=explode, labels=labels, colors=colors,
autopct='%1.1f%%', shadow=True, startangle=140)
plt.axis('equal')
plt.title('Distribution of Broken and Healthy Counts', color = "maroon",
fontsize = 8)
```

3.4.5.5. Based on the amount of impurity content present in the rice sample

- Purity is related to the presence of dockage in the grain. Dockage refers to material other than paddy and includes chaff, stones, weed seeds, soil, rice straw, stalks etc. These impurities generally come from the filed or from the drying floor.
- These unclean materials affects the quality of rice
- So to detect impurities in the sample, first I have imported the images in color format and then the images were converted into HSV color space.
- First I have implemented a method to identify husks and paddy seeds. A mask was created to isolate the impurities.

```
image = cv2.imread('Images/img_01_imp.jpg')
hsv_image = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
lower_color = np.array([0, 50, 50])
upper_color = np.array([20, 255, 255])
color_mask = cv2.inRange(hsv_image, lower_color, upper_color)
```

- Then the impurity area was calculated.

```
contours, _ = cv2.findContours(color_mask, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)
total_impurity_area = 0
for contour in contours:
    total_impurity_area += cv2.contourArea(contour)
```

- After that total area was calculated and impurity percentage was obtained.

```
total_rice_area = image.shape[0] * image.shape[1]

# Calculate the percentage of impurity area
print(f"Percentage of impurity area: {impurity_percentage:.2f}%")
impurity_percentage = (total_impurity_area / total_rice_area) * 100

# Draw contours on the original image
rice_with_contours_imp = img_rgb.copy()
cv2.drawContours(rice_with_contours_imp, contours, -1, (0, 0, 255), 2)
print("**********Analysis based on the impurity content**********")
print(f"Percentage of impurity area: {impurity_percentage:.2f}%")
```

Final Analysis of Quality of Basmati Rice representation using a pie chart.

```
categories = [
    "High Quality Rice (Major Axis Length)",
    "Low Quality Rice (Major Axis Length)",
    "Broken Rice",
    "Chalky Rice",
    "Impurity Content",
]

percentages = [
    (percentage_extra_long_grain + percentage_long_grain),
    percentage_medium_grain,
    broken_count / total_seeds * 100,
    chalky_percentage,
    impurity_percentage,
]

# Create a pie chart
plt.figure(figsize=(10, 8))
plt.pie(percentages, labels=categories, autopct='%1.2f%%', startangle=140)
plt.title("Rice Quality Analysis Summary", fontsize = 16)
plt.axis('equal')
```

CHAPTER 04

RESULTS AND DISCUSSION

4.1.1 GETTING THE COUNT OF RICE GRAINS IN A GIVEN SAMPLE

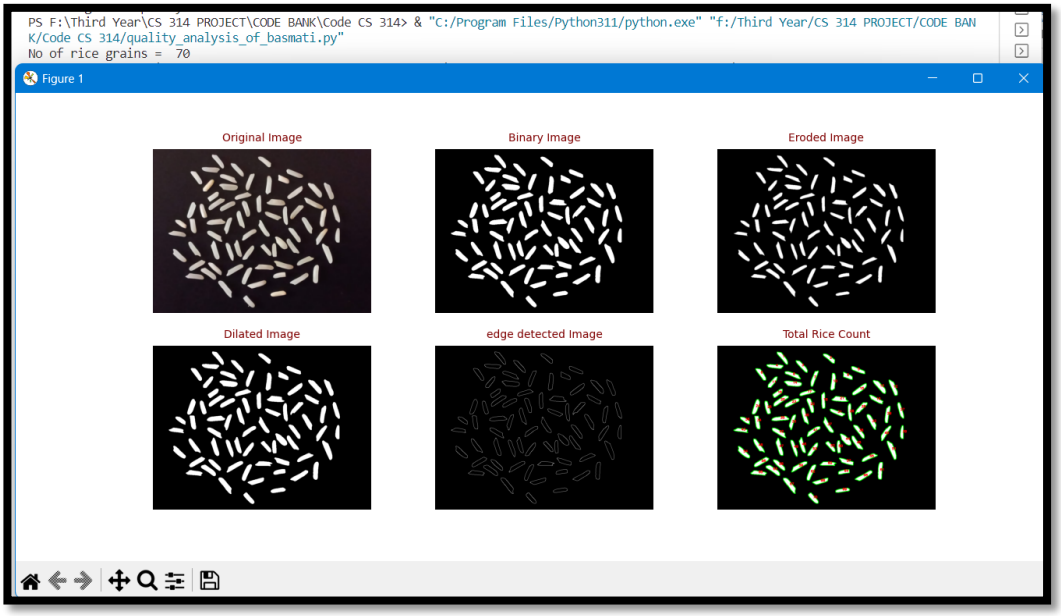


Figure 9: Results of Rice grains after segementation

4.1.2 CALCULATING MAJOR AXIS LENGTH AND MINOR AXIS LENGTH OF EACH RICE GRAIN AND CALCULATING THE ASPECT RATIO OF EACH RICE GRAIN.

```
PS F:\Third Year\CS 314 PROJECT\CODE BANK\Code CS 314> & "C:/Program Files/Python311/python.exe" "f:/Third Year/CS 314 PROJECT/CODE BANK/Code CS 314/quality_analysis
_of_basmati.py"
No of rice grains =  70
Seed Number:  70  |  Major axis length:  67.10263061523438  | Minor axis length:  22.139318466186523  | Aspect Ratio:  3.030925758519645  |
Seed Number:  69  |  Major axis length:  75.09298706054688  | Minor axis length:  16.522178649902344  | Aspect Ratio:  4.5449809405729145  |
Seed Number:  68  |  Major axis length:  85.98969268798828  | Minor axis length:  18.654693603515625  | Aspect Ratio:  4.609547308339755  |
Seed Number:  67  |  Major axis length:  105.47045135498047  | Minor axis length:  22.993438720703125  | Aspect Ratio:  4.586980339744296  |
Seed Number:  66  |  Major axis length:  104.10112762451172  | Minor axis length:  19.942068099975586  | Aspect Ratio:  5.220177120177379  |
Seed Number:  65  |  Major axis length:  79.74593353271484  | Minor axis length:  19.325462341308594  | Aspect Ratio:  4.1264696349466465  |
Seed Number:  64  |  Major axis length:  65.9420166015625  | Minor axis length:  18.172967910766 | Aspect Ratio:  3.628577177121138  |
Seed Number:  63  |  Major axis length:  92.07289123535156  | Minor axis length:  21.56730842590332  | Aspect Ratio:  4.26909512383881  |
Seed Number:  62  |  Major axis length:  99.94580841064453  | Minor axis length:  19.637752532958984  | Aspect Ratio:  5.089472853011091  |
Seed Number:  61  |  Major axis length:  94.45138549804688  | Minor axis length:  21.191431045532227  | Aspect Ratio:  4.457055556800634  |
Seed Number:  60  |  Major axis length:  75.51390838623047  | Minor axis length:  21.575658798217773  | Aspect Ratio:  3.4999584064829663  |
Seed Number:  59  |  Major axis length:  97.468994140625  | Minor axis length:  19.318952560424805  | Aspect Ratio:  5.045252522659632  |
Seed Number:  58  |  Major axis length:  76.59907531738281  | Minor axis length:  19.34750747680664  | Aspect Ratio:  3.959118527759096  |
Seed Number:  57  |  Major axis length:  84.2599105834961  | Minor axis length:  19.830625534057617  | Aspect Ratio:  4.248978956250573  |
Seed Number:  56  |  Major axis length:  88.3000259399414  | Minor axis length:  19.925600051879883  | Aspect Ratio:  4.431486414965492  |
Seed Number:  55  |  Major axis length:  85.71647644042969  | Minor axis length:  18.86551284790039  | Aspect Ratio:  4.543554004150455  |
Seed Number:  54  |  Major axis length:  75.98128509521484  | Minor axis length:  24.333112716674805  | Aspect Ratio:  3.122546875934495  |
Seed Number:  53  |  Major axis length:  85.31256103515625  | Minor axis length:  18.615537643432617  | Aspect Ratio:  4.582868497770931  |
Seed Number:  52  |  Major axis length:  65.42216491699219  | Minor axis length:  18.99732208251953  | Aspect Ratio:  3.443757211296148  |
Seed Number:  51  |  Major axis length:  104.14959716796875  | Minor axis length:  22.761186599731445  | Aspect Ratio:  4.575754287309326  |
Seed Number:  50  |  Major axis length:  68.2221450805664  | Minor axis length:  16.309720993041992  | Aspect Ratio:  4.182913068204609  |
Seed Number:  49  |  Major axis length:  95.64606475830078  | Minor axis length:  16.08213233947754  | Aspect Ratio:  5.947349688418746  |
Seed Number:  48  |  Major axis length:  81.81828308105469  | Minor axis length:  20.896865844726562  | Aspect Ratio:  3.9153375290343826  |
Seed Number:  47  |  Major axis length:  91.80934906005586  | Minor axis length:  20.49503517150879  | Aspect Ratio:  4.479589729501296  |
Seed Number:  46  |  Major axis length:  80.6683578491211  | Minor axis length:  18.870637893676758  | Aspect Ratio:  4.274808212824206  |
Seed Number:  45  |  Major axis length:  92.579833984375  | Minor axis length:  20.50023078918457  | Aspect Ratio:  4.516038621048984  |
Seed Number:  44  |  Major axis length:  68.52198791503906  | Minor axis length:  27.516725540161133  | Aspect Ratio:  2.4901941117605015  |
Seed Number:  43  |  Major axis length:  69.4209213256836  | Minor axis length:  17.59610366821289  | Aspect Ratio:  3.9452439377867212  |
Seed Number:  42  |  Major axis length:  85.86956024169922  | Minor axis length:  20.639942169189453  | Aspect Ratio:  4.1603585677619845  |
Seed Number:  41  |  Major axis length:  84.96141052246094  | Minor axis length:  19.092618942260742  | Aspect Ratio:  4.449961044076686  |
Seed Number:  40  |  Major axis length:  101.69005584716797  | Minor axis length:  19.27362060546875  | Aspect Ratio:  5.276126262354368  |
Seed Number:  39  |  Major axis length:  65.08169555664062  | Minor axis length:  21.2266845703125  | Aspect Ratio:  3.0660320664331846  |
Seed Number:  38  |  Major axis length:  92.80652618408203  | Minor axis length:  18.625837326049805  | Aspect Ratio:  4.982676728003217  |
Seed Number:  37  |  Major axis length:  97.65718078613281  | Minor axis length:  20.293851852416992  | Aspect Ratio:  4.812155991692719  |
Seed Number:  36  |  Major axis length:  57.6692008972168  | Minor axis length:  18.213134765625  | Aspect Ratio:  3.166352285827268  |
Seed Number:  35  |  Major axis length:  86.21613311767578  | Minor axis length:  20.60596466064453  | Aspect Ratio:  4.184037706438493  |
Seed Number:  34  |  Major axis length:  88.33522033691406  | Minor axis length:  20.008472442626953  | Aspect Ratio:  4.414890771407452  |
Seed Number:  33  |  Major axis length:  81.51994323730469  | Minor axis length:  17.86493682861328  | Aspect Ratio:  4.563125188707004  |
Seed Number:  32  |  Major axis length:  91.6337661743164  | Minor axis length:  20.41808319091797  | Aspect Ratio:  4.487873093546578  |
Seed Number:  31  |  Major axis length:  87.21427917480469  | Minor axis length:  17.26925277709961  | Aspect Ratio:  5.050263627529831  |
Seed Number:  30  |  Major axis length:  87.62898254394531  | Minor axis length:  19.53009796142578  | Aspect Ratio:  4.486868561387805  |
```

Figure 10: Measurements of each rice grain in a sample

4.1.3 GRADING THE RICE BASED ON MAJOR AXIS LENGTH

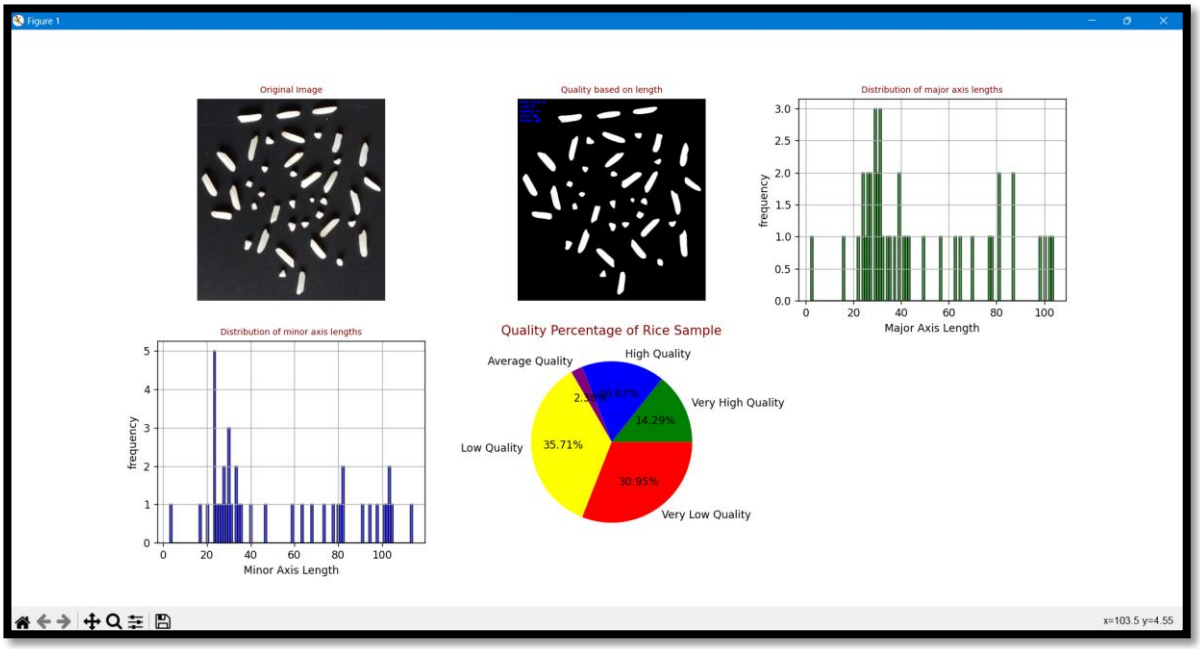


Figure 11: Analysis based on Major Axis Length

```
********************Counts ********************

Count of Extra Long Grain: 6
Count of Long Grain: 7
Count of Medium Grain: 1
Count of Short Grain: 15
Count of Broken Grain: 13

**********Analysis based on the major axis length of the rice granules******
14.29% of Very High Quality Rice
16.67% of High Quality Rice
2.38% of  Average Quality Rice
35.71% of Low Quality Rice
30.95% of Very Low Quality Rice
```

Figure 11.1: Analysis based on Major Axis Length Percentages

4.1.4 GRADING THE RICE BASED ON ASPECT RATIO (GRAIN SHAPE)

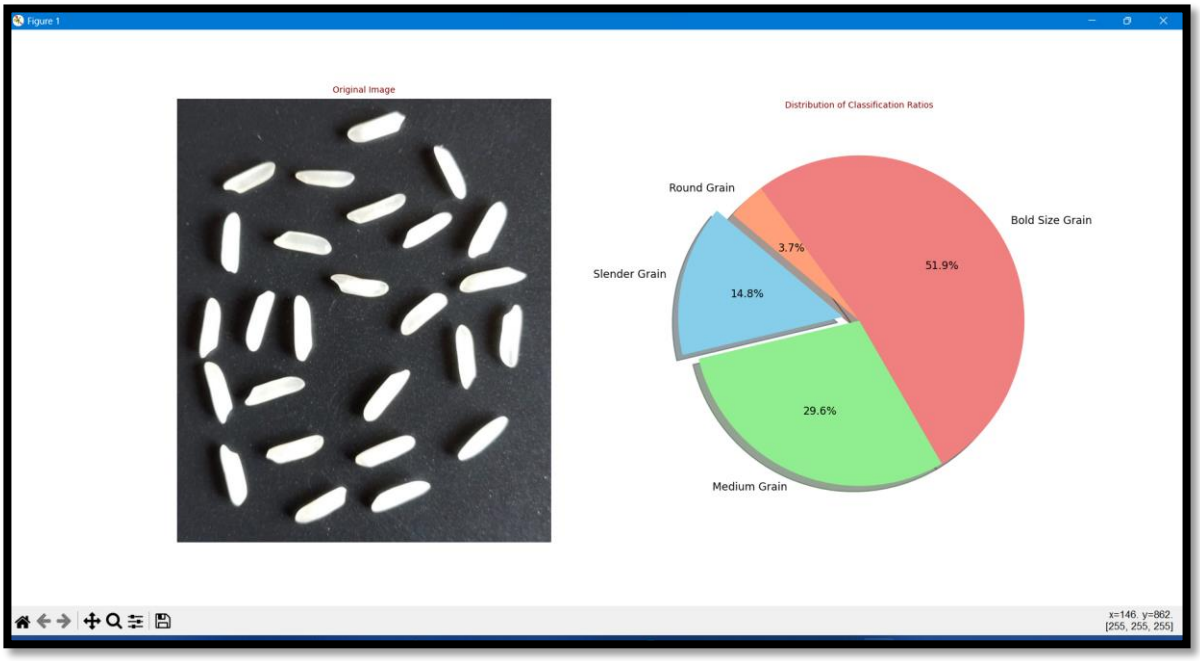


Figure 12: Analysis based on Aspect Ratio

```
******************Count*******************
No of rice grains =  27
Count of Slender Grain: 4
Count of Medium Grain: 8
Count of Bold Size Grain: 14
Count of Round Grain: 1

**********Percentage of each category *********
Percentage of Slender Grain: 14.814814814814813
Percentage of Medium Grain: 29.629629629629626
Percentage of Bold Size Grain: 51.85185185185185
Percentage of Round Grain: 3.7037037037037033

**********Analysis based on the aspect ratio of the rice granules**********
14.81% of Very High Quality Rice
29.63% of High Quality Rice
51.85% of Average Quality Rice
3.70% of Low Quality Rice
```

Figure 12.1: Analysis based on Aspect Ratio Percentages

4.1.5 GRADING THE RICE BASED ON CHALKY CONTENT

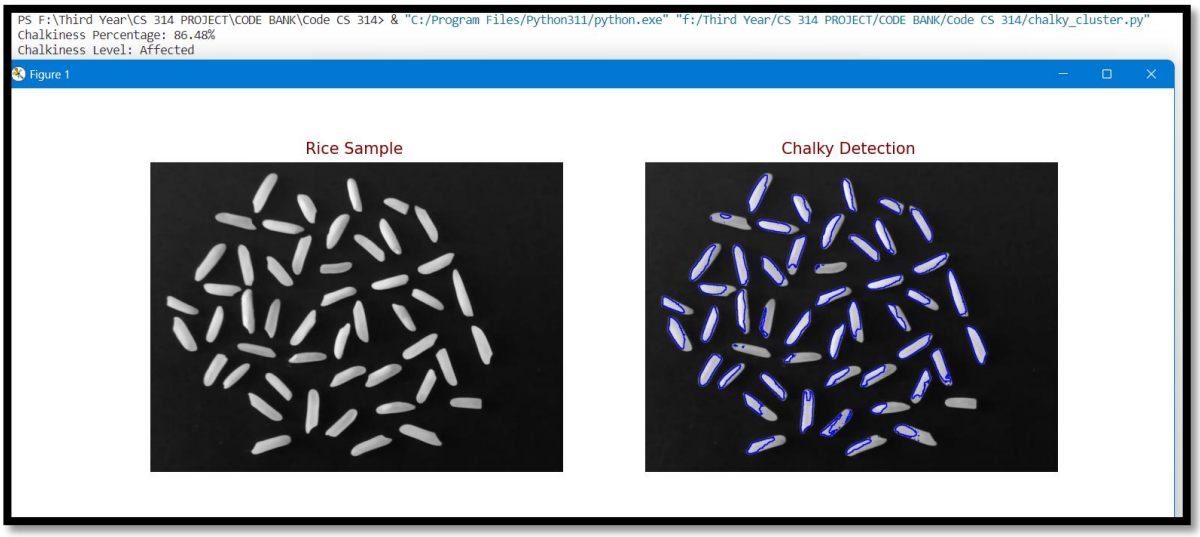


Figure 13: Analysis based on Chalky Content

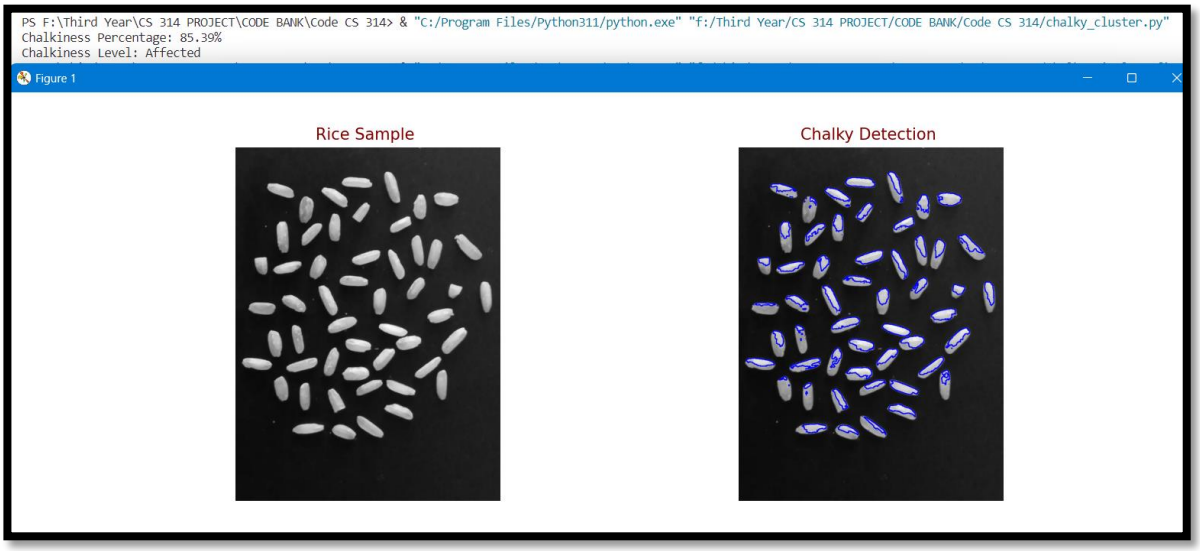


Figure 13.1: Analysis based on Chalky Content

4.1.6 GRADING THE RICE BASED ON BROKEN RICE GRAINS



Figure 14: Analysis based on Broken Seed Content

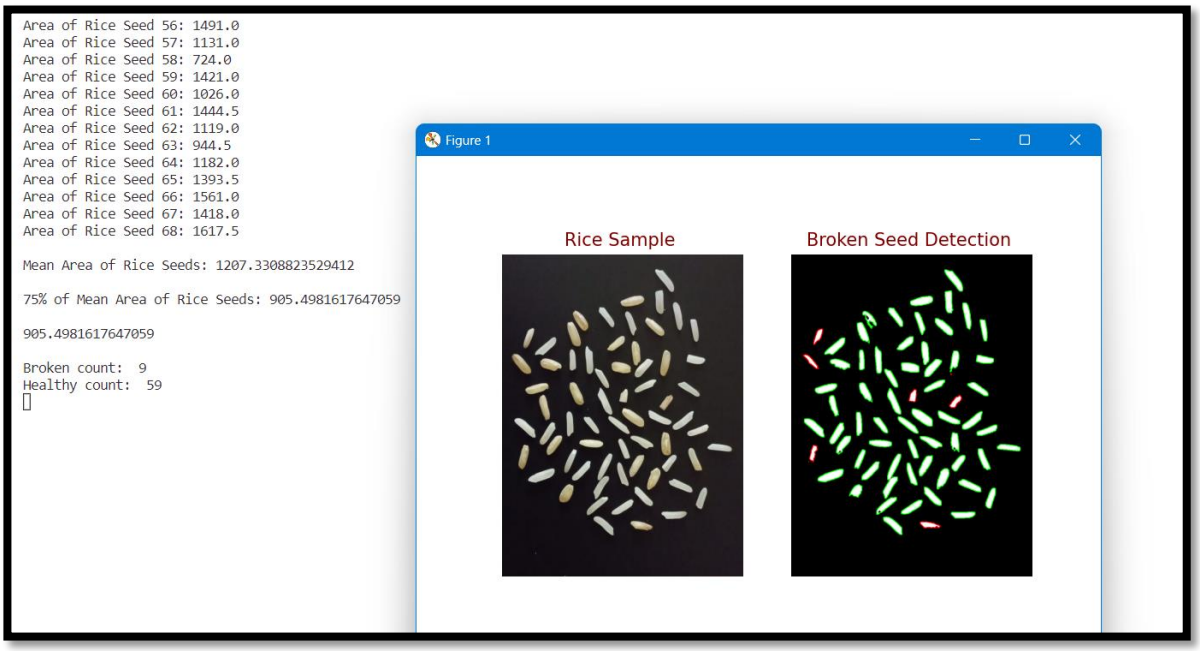


Figure 14.1: Analysis based on Broken Seed Content

4.1.7 GRADING THE RICE BASED ON IMPURITY DETECTION



Figure 15: Impurity Detection paddy seeds

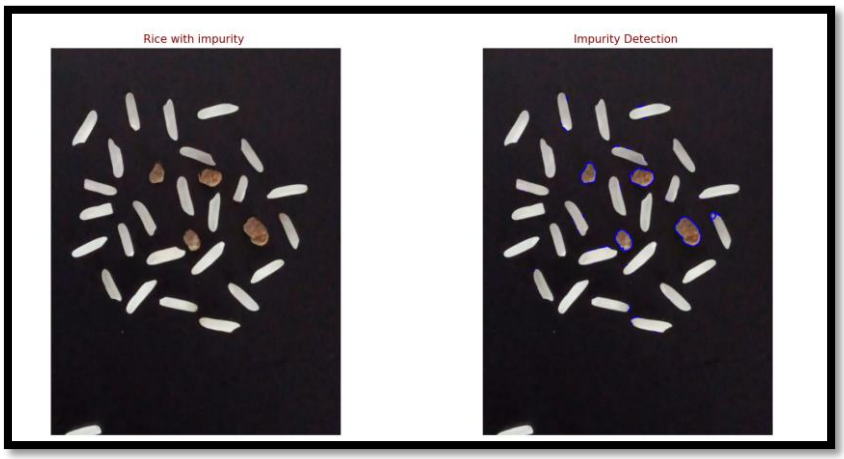


Figure 15.1: Impurity Detection rocks

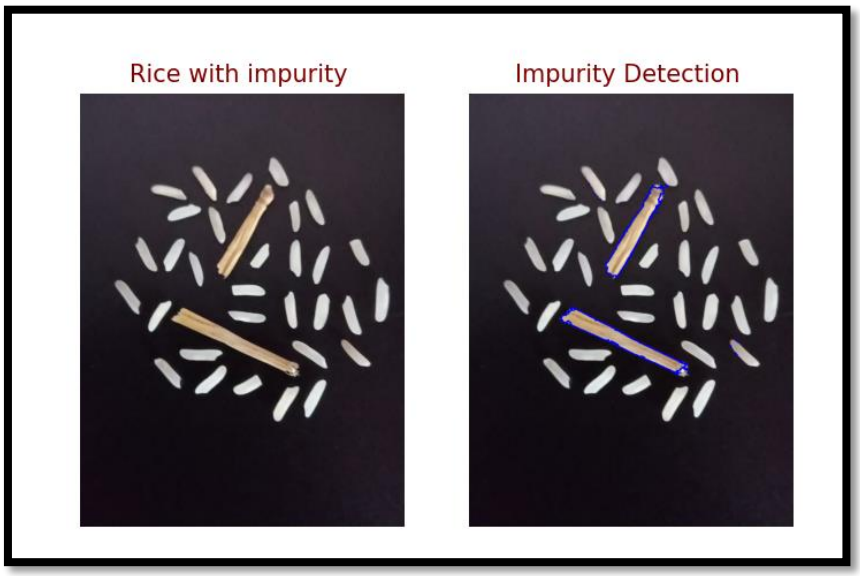


Figure 15.2: Impurity Detection

4.1.8 FINAL ANALYSIS OF RICE IN A SAMPLE

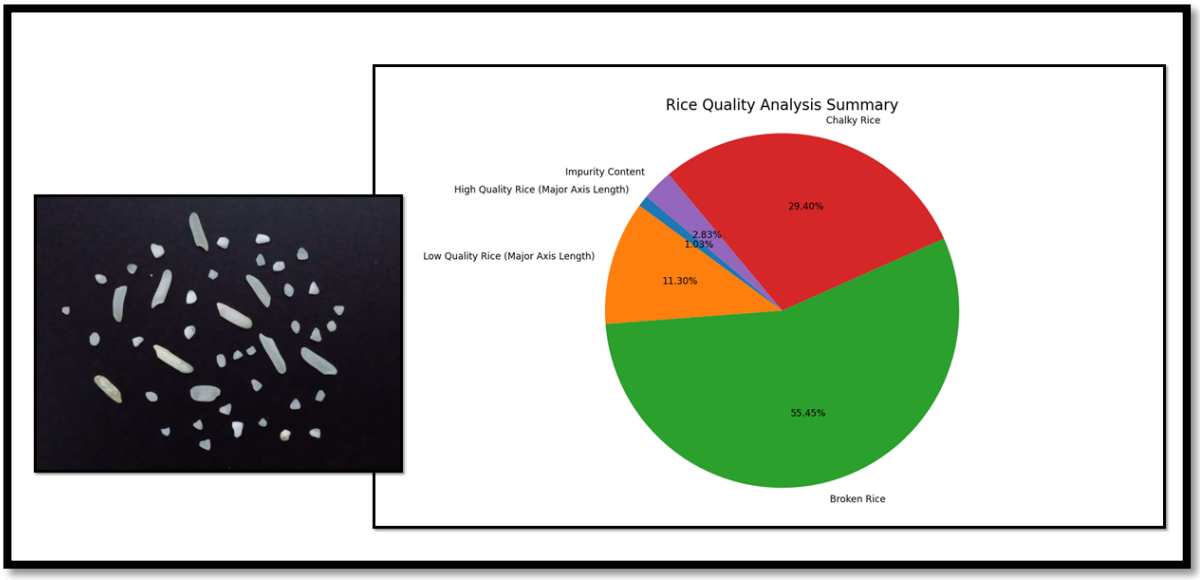


Figure 16: Analysis Rice Sample 01

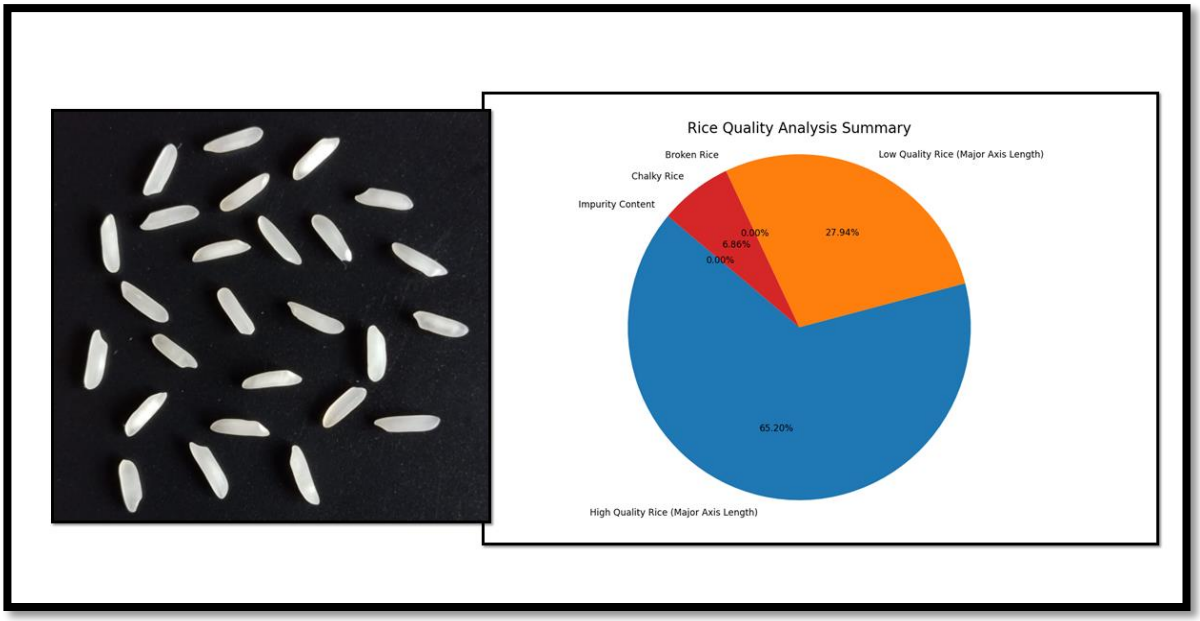


Figure 16.1: Analysis Rice Sample 02

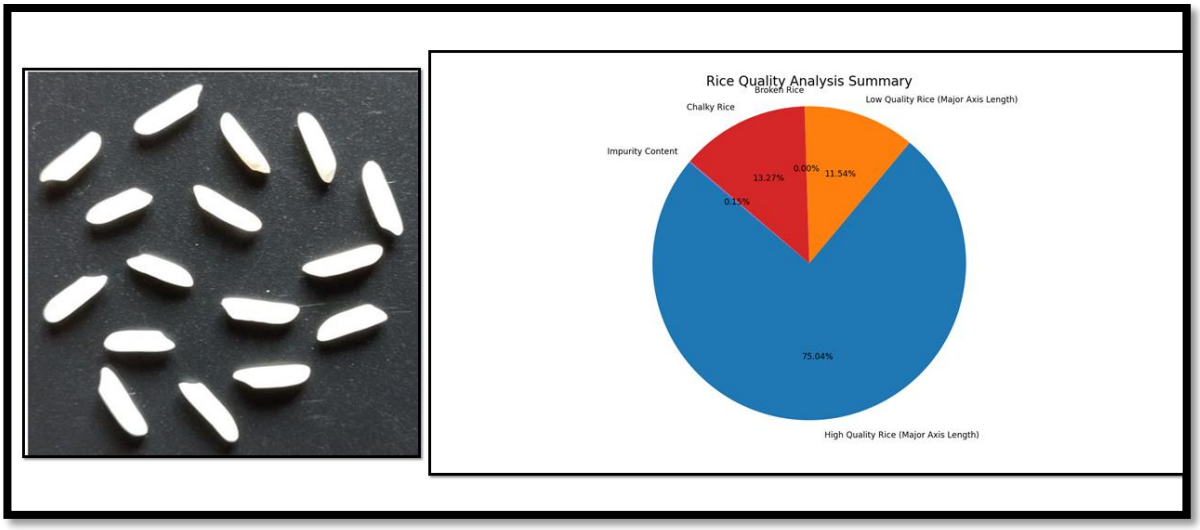


Figure 16.2: Analysis Rice Sample 03

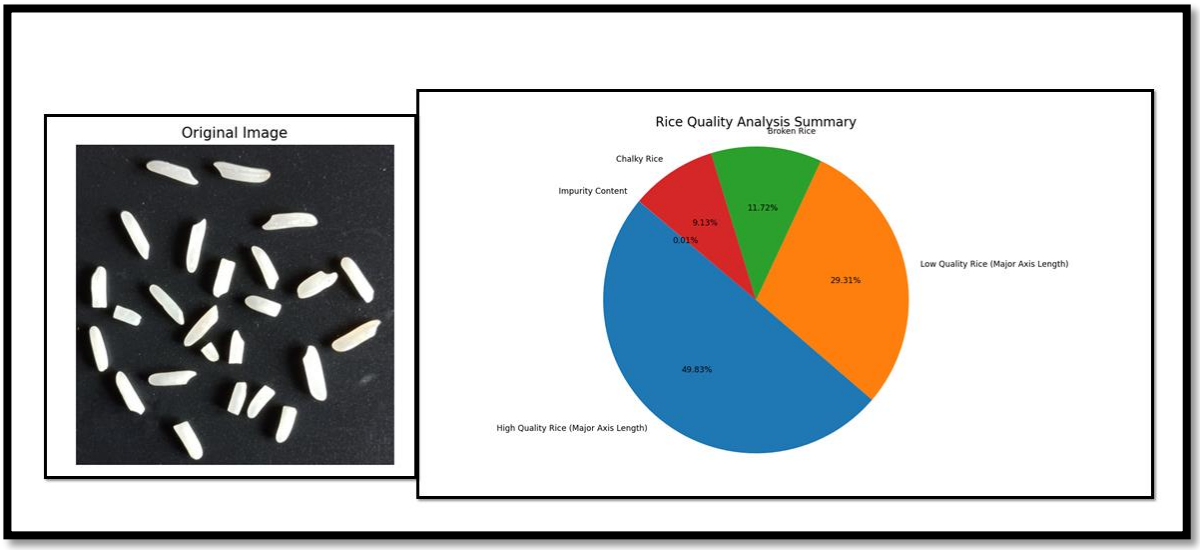


Figure 16.3 Analysis Rice Sample 04

CHAPTER 05

CONCLUSIONS

Rice is one of the main food source of more than 80% of the population. The demand for rice is increasing day by day tremendously. So quality analyzing and categorization should be done fastly to meet the needs of people. Use of traditional methods and manual inspection methods does not provide a good analysis. These manual inspection methods are very time consuming, need lot of manpower, costly and inefficient. This project of quality analysis of basmati rice using image processing techniques mainly aims in solving the above problems leading to very efficient accurate quality analysis of rice.

The system identifies number of seeds in a sample, and calculates major axis length, minor axis length, area of each rice seed and then based on the measurements the system is capable of classifying rice based on major axis length, aspect ratio, chalky content, broken seed content, and amount of impurity content in a given sample.

So using this system we can overcome the problems and difficulties occur in manual inspection methods. The setup used is simple. So as of conclusion we can say that this method is much more accurate than the human visual inspection which provides very accurate results which helps in grading and categorizing Basmati Rice.

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